

Loan Approval Prediction Report

1. Dataset Description

In this project, we worked with a Loan Approval dataset. The goal was to predict whether a loan application would be approved or not based on different applicant features.

The dataset contains both numerical and categorical features such as:

- Loan id
- no_of_dependents
- education
- self_employed
- income_annum
- loan_amount
- loan_term
- cibil_score
- residential_assets_value
- commercial_assets_value
- luxury_assets_value
- bank_asset_value

The target variable is **Loan_Status**, which is binary (approved or not approved).

Before training the models, several preprocessing steps were applied. Missing values were handled, categorical features were encoded, and the data was split into training and testing sets. Feature scaling was applied where necessary, especially for Logistic Regression.

To solve the classification problem, we implemented and compared different models:

2. Methodology

- Logistic Regression with L2 regularization

- Logistic Regression with L1 regularization
- Decision Tree using Gini
- Decision Tree using Entropy
- Manually tuned Decision Tree
- GridSearchCV for automatic hyperparameter tuning

Logistic Regression

Logistic Regression is a linear classification algorithm. We tested two types of regularization:

- **L2 (Ridge)**: helps reduce large coefficients.
- **L1 (Lasso)**: can eliminate less important features by shrinking some coefficients to zero.

Regularization helps prevent overfitting and improves the model's ability to generalize.

Decision Tree

Decision Trees are non-linear models that split the data based on feature values. We tested both Gini and Entropy criteria to see which performs better.

Decision Trees can model complex relationships but may overfit if not controlled properly. That's why we also tried manual tuning and GridSearchCV to optimize parameters such as max depth and minimum samples.

3. GridSearchCV

GridSearchCV was used to find the best hyperparameters automatically using cross-validation. This helped improve model performance and avoid relying on trial and error.

4. Results and Analysis

Below is a summary of the results:

Logistic Regression (L2) achieved about **79.7% test accuracy**, with lower recall (0.64). This suggests that the model missed a noticeable number of positive cases. It seems to underfit the data.

Logistic Regression (L1) performed much better, with around **93.6% test accuracy** and a strong F1-score (0.9137). L1 regularization likely helped by removing less important features.

Decision Tree (Gini) and Decision Tree (Entropy) both achieved perfect training accuracy (100%), which indicates overfitting. However, their test accuracy was still very high (around 99%).

The manually tuned Decision Tree improved generalization and achieved **99.61% test accuracy** with an F1-score of about **0.9948**.

The GridSearch Decision Tree gave the best overall performance, also reaching **99.61% test accuracy** with excellent precision and recall.

Overall, Decision Tree models clearly outperformed Logistic Regression in this dataset.

5. Conclusion

Based on the results, the **GridSearch-tuned Decision Tree** performed the best.

It achieved the highest test accuracy and F1-score while maintaining a very small gap between training and testing accuracy. This means it generalized well and did not suffer from severe overfitting after tuning.

Although Logistic Regression (especially L1) gave good results, it could not match the performance of the Decision Tree. This suggests that the relationships in the dataset are likely non-linear, which tree-based models can handle better than linear models.

In conclusion, for this loan approval dataset, the tuned Decision Tree is the most suitable model due to its high accuracy, strong balance between precision and recall, and better handling of feature interactions.